

PVsyst - Simulation report

Grid-Connected System

Project: PV_toitures

Variant: simulation_PV

Building system

System power: 412 kWp

Suissi - Morocco

**PVsyst V8.1.2**

VC0, Simulation date:

03/06/26 14:51

with V8.1.2

Project summary**Geographical Site**

Suissi

Maroc

Situation

Latitude 33.9823 °(N)
Longitude -6.8557 °(W)
Altitude 46 m
Time zone UTC

Project settings

Albedo 0.20

Weather data

Suissi

Meteonorm 9.0 dll, Sat=100 % - Synthétique

System summary**Grid-Connected System****Orientation #1****Fixed plane**

Tilt/Azimuth 30 / -51 °

System information**PV Array**

Nb. of modules

936 units

Pnom total

412 kWp

Building system**Near Shadings**

According to strings : Fast (table)

Electrical effect 100 %

User's needs

Unlimited load (grid)

Inverters

Nb. of units

1.1 unit

Total power

379 kWac

Pnom ratio

1.09

Results summary

Simulation step Hourly

Produced Energy 573984 kWh/year Specific production 1394 kWh/kWp/year Perf. Ratio PR 71.95 %

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General parameters**Grid-Connected System****Simulation step**

Hourly

Orientation #1**Fixed plane**

Tilt/Azimuth 30 / -51 °

Building system**Sheds configuration**

Nb. of sheds 10 units

Set of tables

Shading limit angle

Limit profile angle °

Sizes

Sheds spacing 0.00 m

Sensitive width 0.00 m

Models used

Transposition Perez

Diffuse Perez, Meteonorm

Circumsolar separate

Horizon

Free Horizon

Near Shadings

According to strings : Fast (table)

Electrical effect 100 %

User's needs

Unlimited load (grid)

PV Array Characteristics**PV module**

Manufacturer Generic

Model Mono 440 Wp Twin 144 half-cells

(Original PVsyst database)

Unit Nom. Power 440 Wp

Number of PV modules 936 units

Nominal (STC) 412 kWp

Modules 39 string x 24 In series

At operating cond. (50°C)

Pmpp 374 kWp

U mpp 899 V

I mpp 416 A

Total PV power

Nominal (STC) 412 kWp

Total 936 modules

Module area 2083 m²Cell area 1857 m²**Inverter**

Manufacturer Generic

Model 350 kWac inverter, 12 MPPT

(Original PVsyst database)

Unit Nom. Power 350 kWac

Number of inverters 13 * MPPT 8% 1.1 unit

Total power 379 kWac

Operating voltage 480-1500 V

Pnom ratio (DC:AC) 1.09

No power sharing between MPPTs

Total inverter power

Total power 379 kWac

Nb. of inverters 2 units

0.9 unused

Pnom ratio 1.09

Array losses**Thermal Loss factor**

Module temperature according to irradiance

Uc (const) 20.0 W/m²KUv (wind) 0.0 W/m²K/m/s**DC wiring losses**

Global array res. 36 mΩ

Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.38 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

Strings Mismatch loss

Loss Fraction 0.10 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

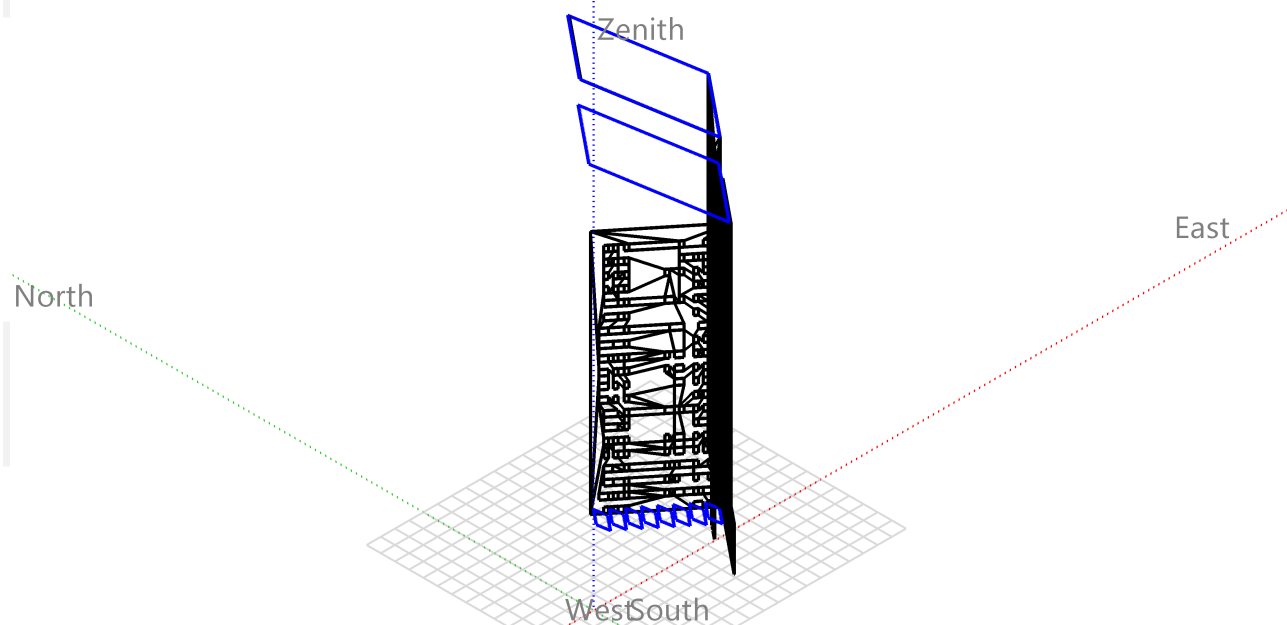


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Near shadings parameter

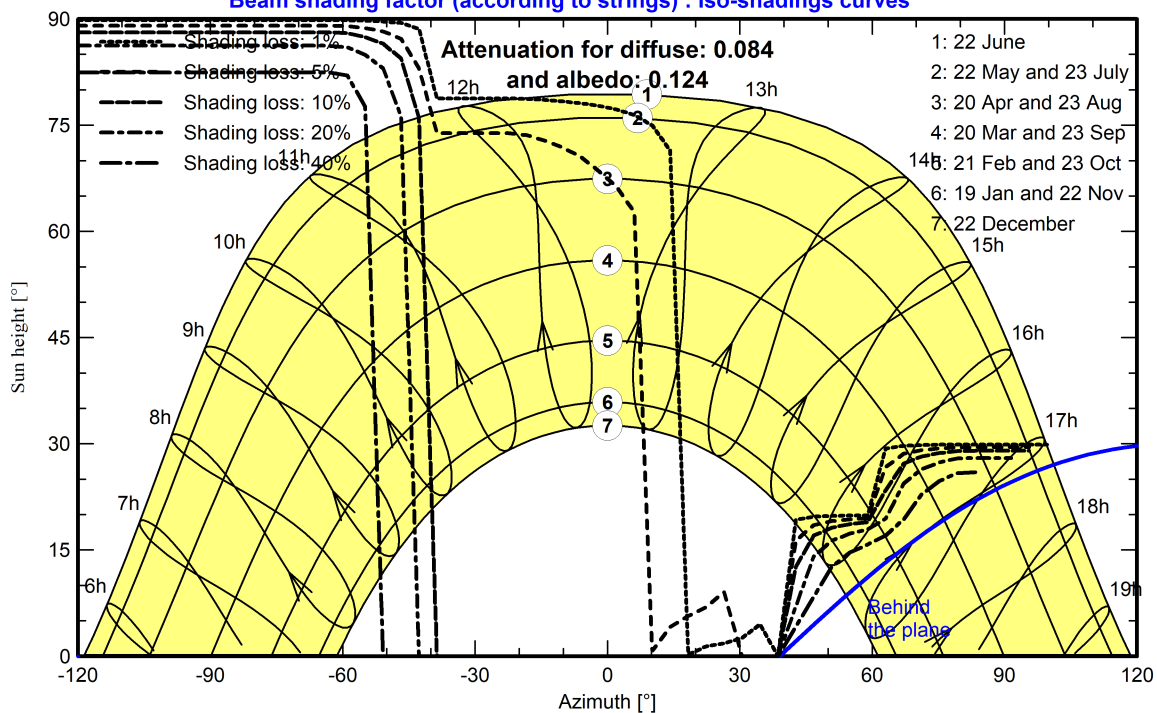
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 30°/-51°

Beam shading factor (according to strings) : Iso-shadings curves





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Main results

Simulation step

Hourly

System Production

Produced Energy 573984 kWh/year

Specific production

1394 kWh/kWp/year

Perf. Ratio PR

71.95 %

Economic evaluation

Investment

Global 240 837.70 EUR

Specific 0.58 EUR/Wp

Yearly cost

Annuities

Run. costs

Payback period

0.00 EUR/yr

5 003.31 EUR/yr

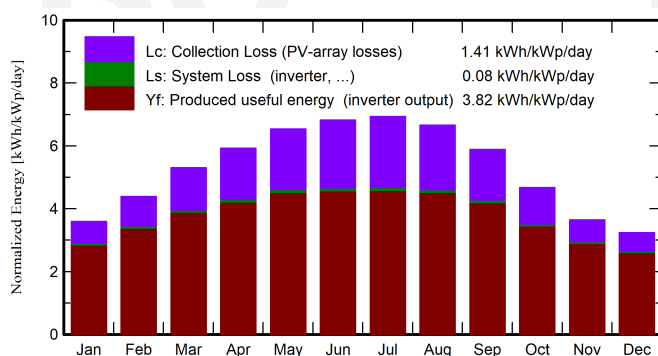
4.7 years

LCOE

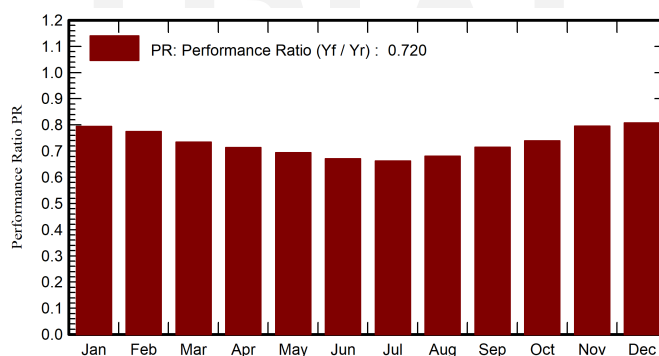
Energy cost

0.05 EUR/kWh

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	86.5	31.72	11.84	111.6	104.2	37282	36509	0.794
February	102.7	39.92	12.64	122.9	114.9	39954	39171	0.774
March	148.8	59.00	14.44	164.6	152.8	50741	49754	0.734
April	177.6	66.53	16.14	177.8	164.3	53271	52242	0.713
May	209.8	81.23	18.64	202.6	185.6	59015	57878	0.694
June	219.6	80.68	21.05	204.8	185.6	57728	56598	0.671
July	228.4	75.44	22.55	215.1	195.9	59809	58639	0.662
August	208.7	69.73	23.15	206.4	190.1	58935	57834	0.680
September	165.7	63.24	21.75	176.4	164.0	52928	51916	0.714
October	127.1	46.13	19.45	145.0	135.7	45050	44159	0.740
November	88.0	42.06	15.44	109.4	102.0	36581	35859	0.796
December	77.2	33.37	13.14	100.5	93.7	34118	33427	0.808
Year	1840.0	689.05	17.55	1937.0	1788.9	585413	573984	0.720

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

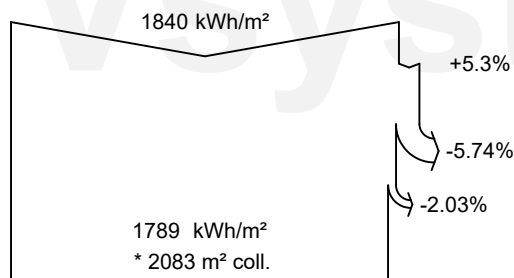
PR Performance Ratio



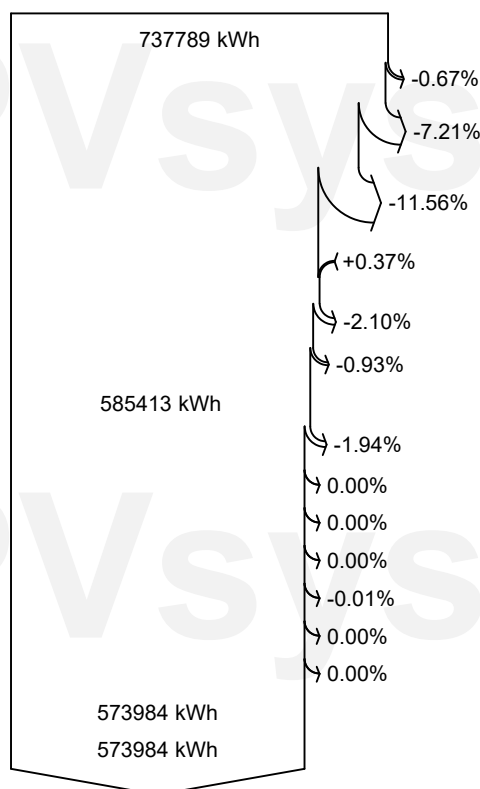
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Loss diagram



efficiency at STC = 19.80%



Global horizontal irradiation

Global incident in coll. plane

Near Shadings: irradiance loss

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Shadings: Electrical Loss acc. to strings

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

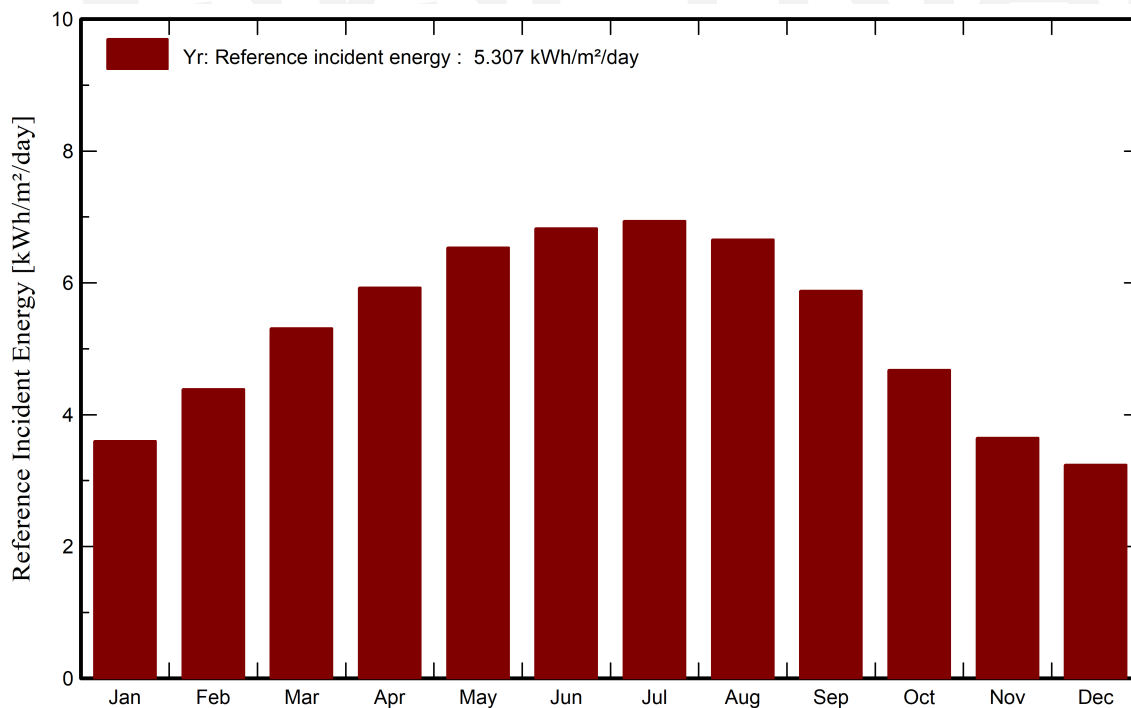


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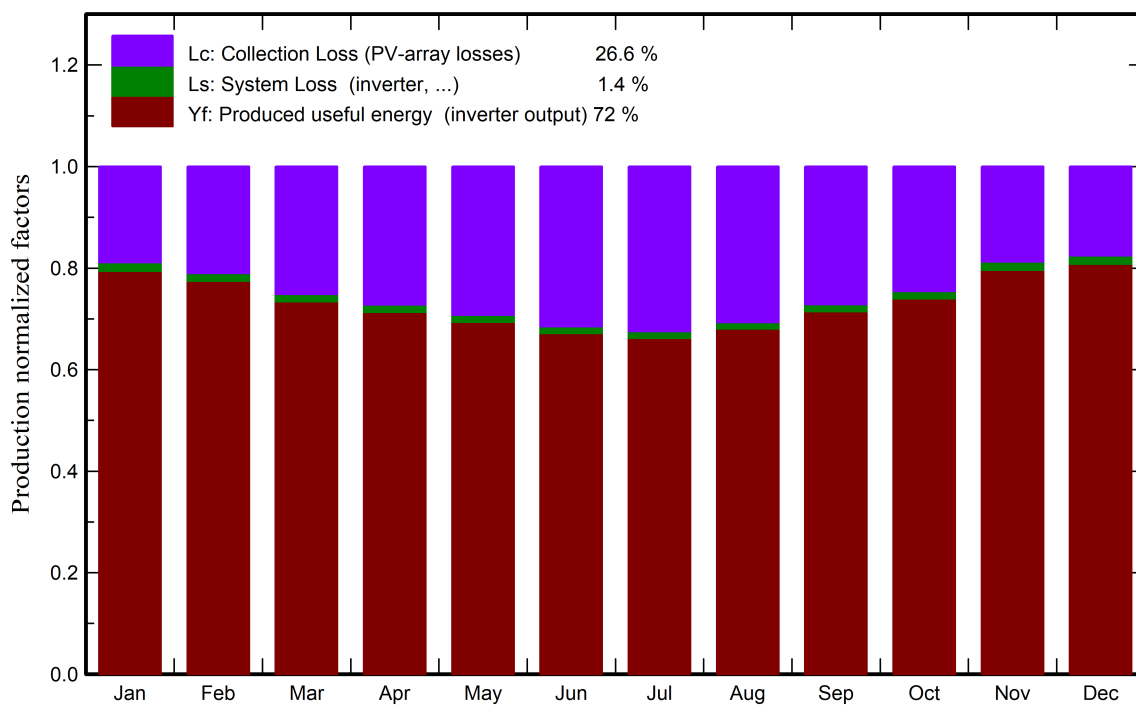
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Predef. graphs

Reference Incident Energy in Collector Plane



Normalized Production and Loss Factors



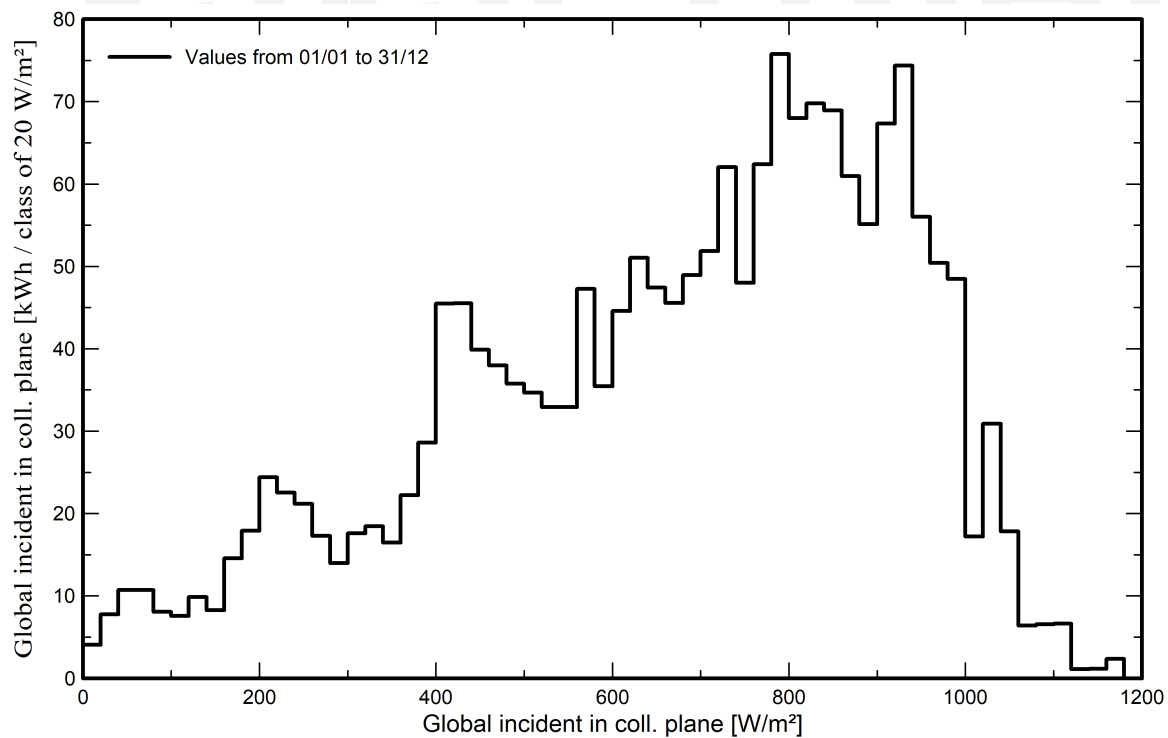


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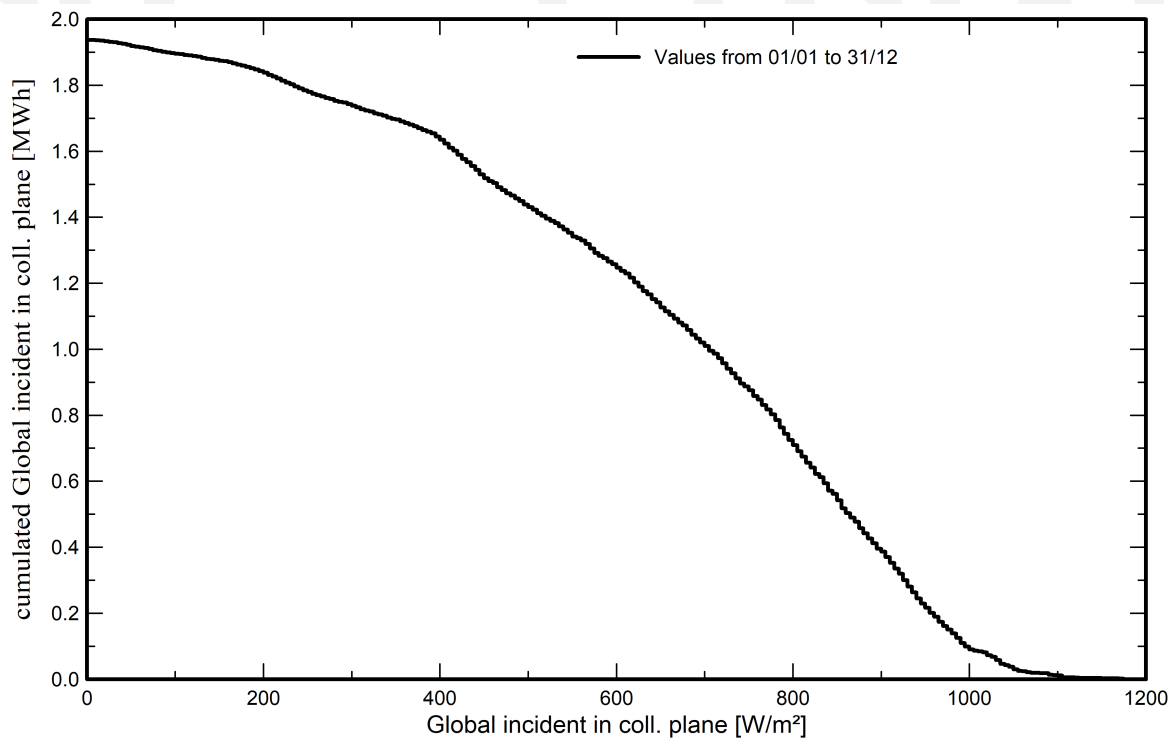
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Predef. graphs

Incident Irradiation Distribution



Incident Irradiation cumulative distribution



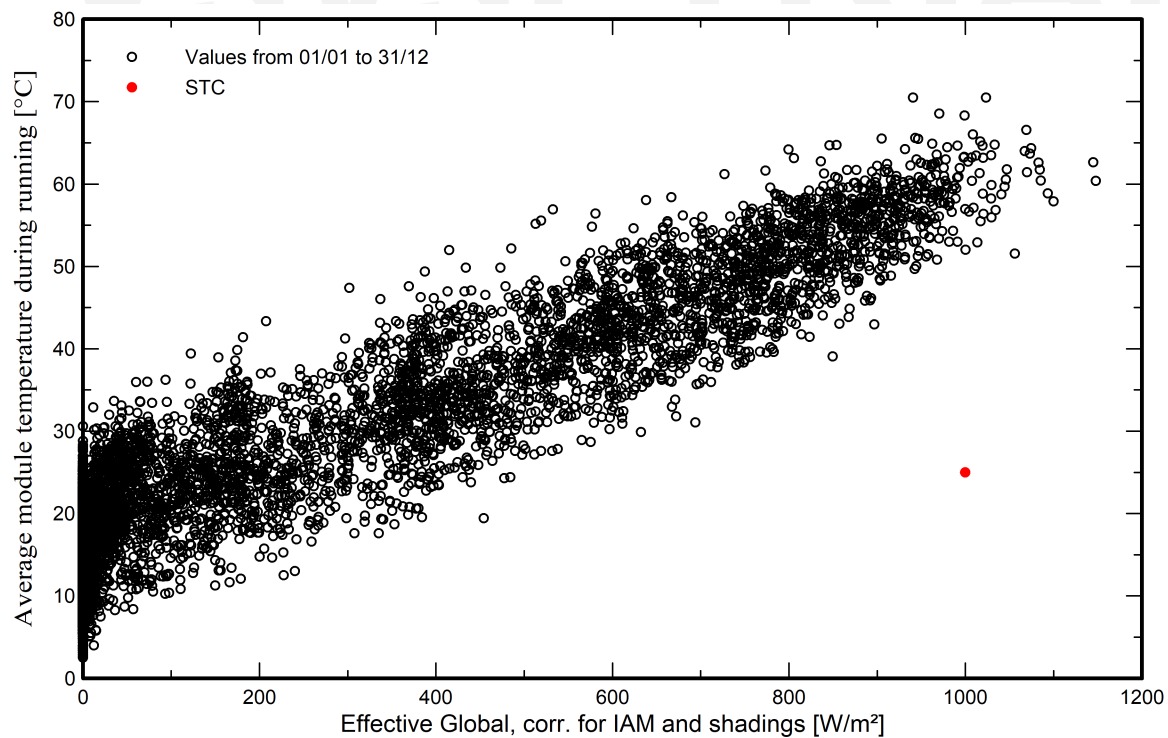


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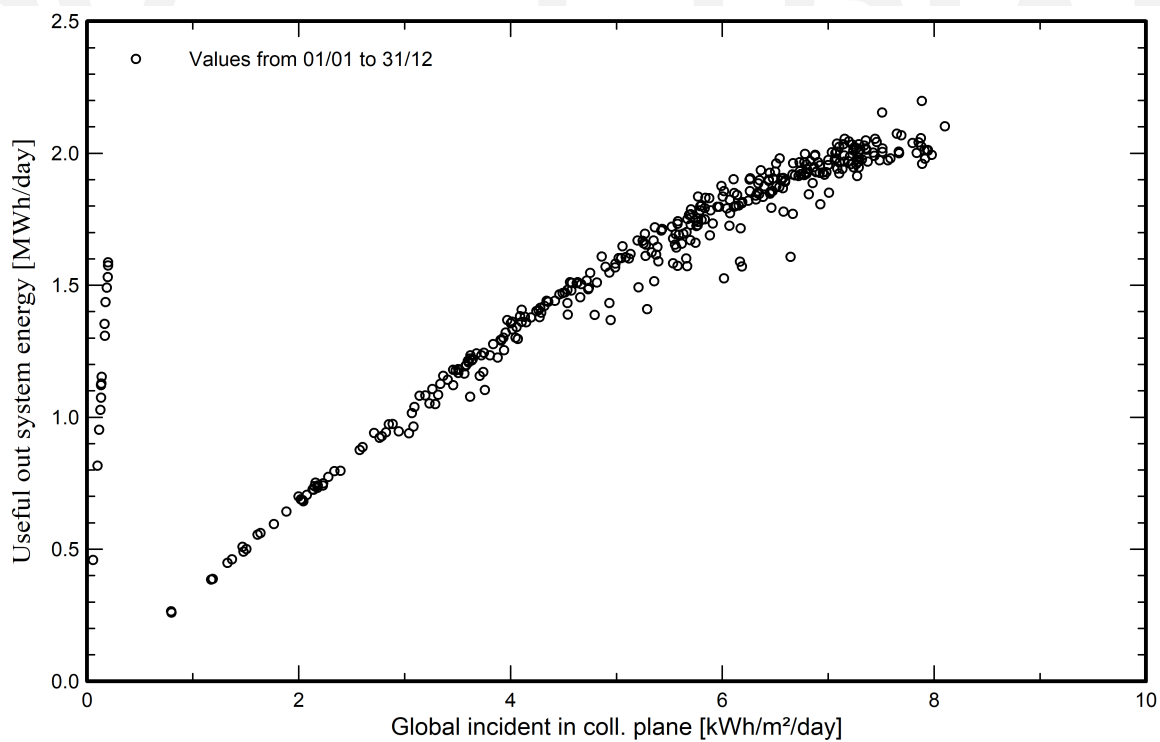
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Predef. graphs

Array Temperature vs. Effective Irradiance



Daily Input/Output diagram

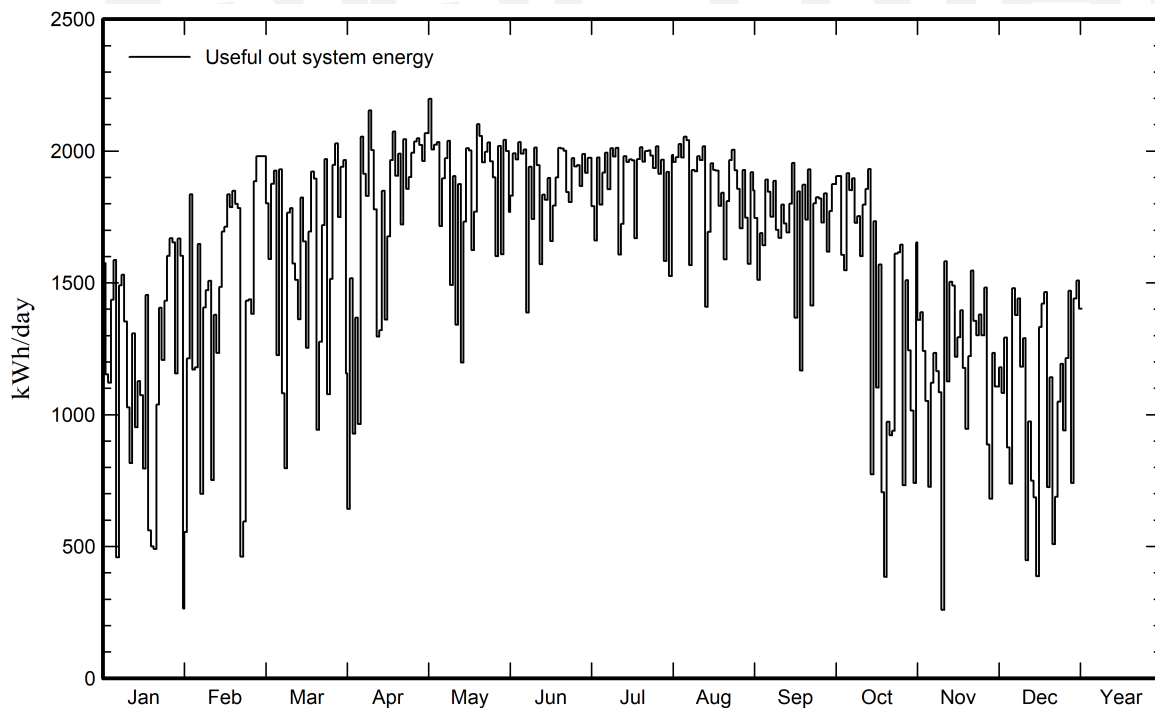




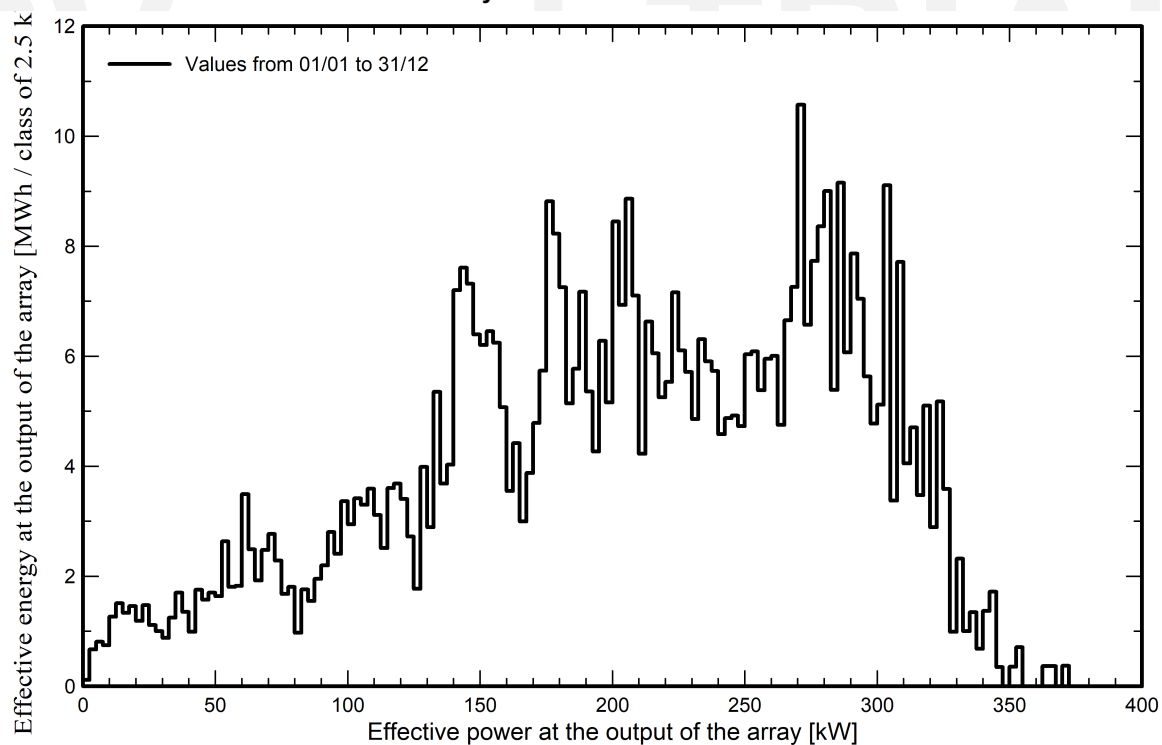
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Predef. graphs
Daily System Output Energy



Array Power Distribution



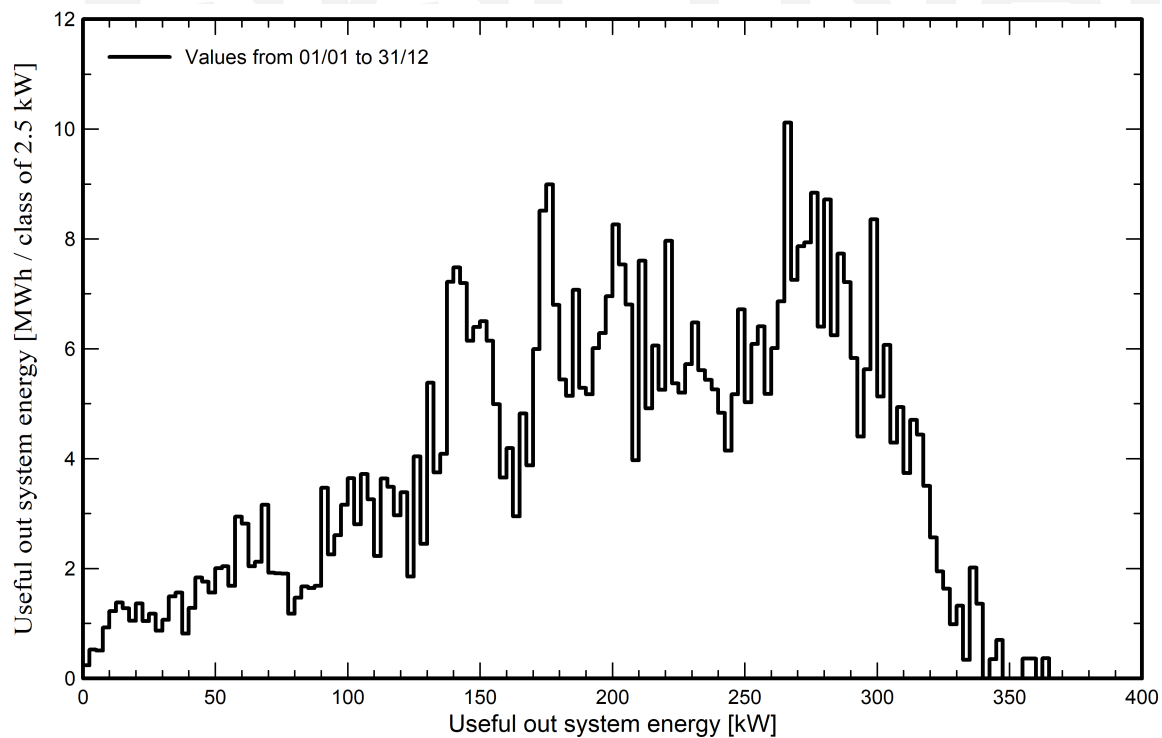


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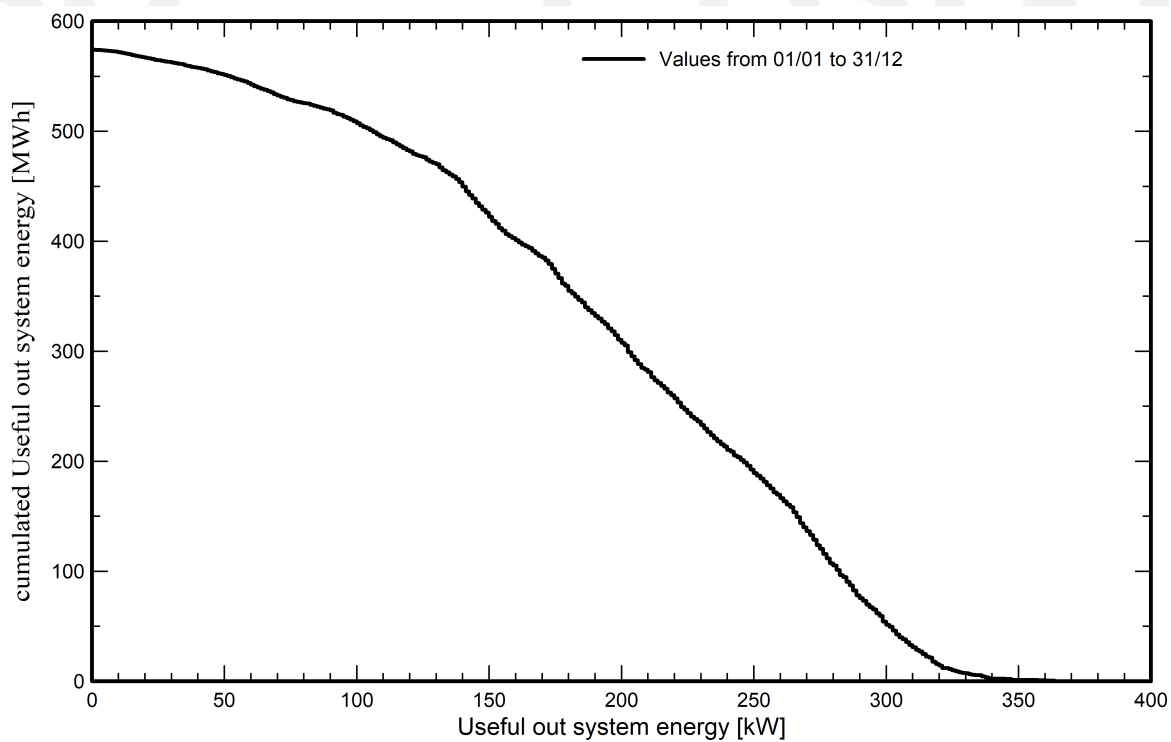
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Predef. graphs

System Output Power Distribution



System Output Power cumulative distribution



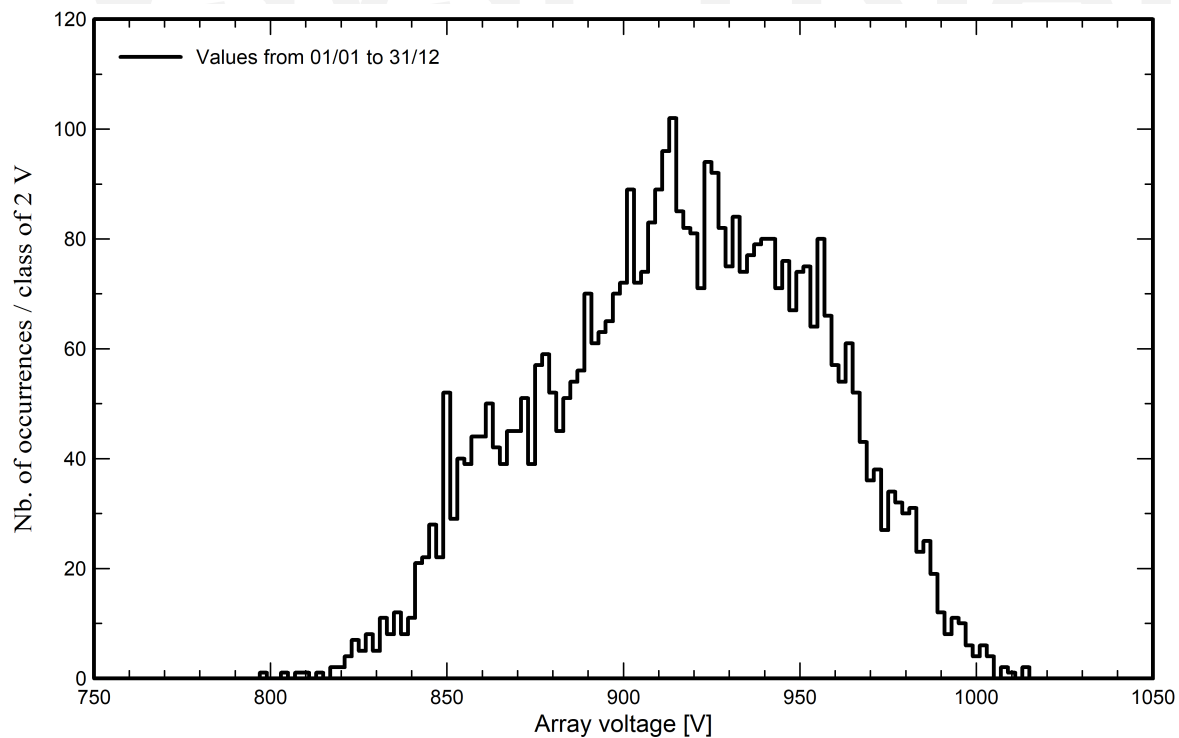


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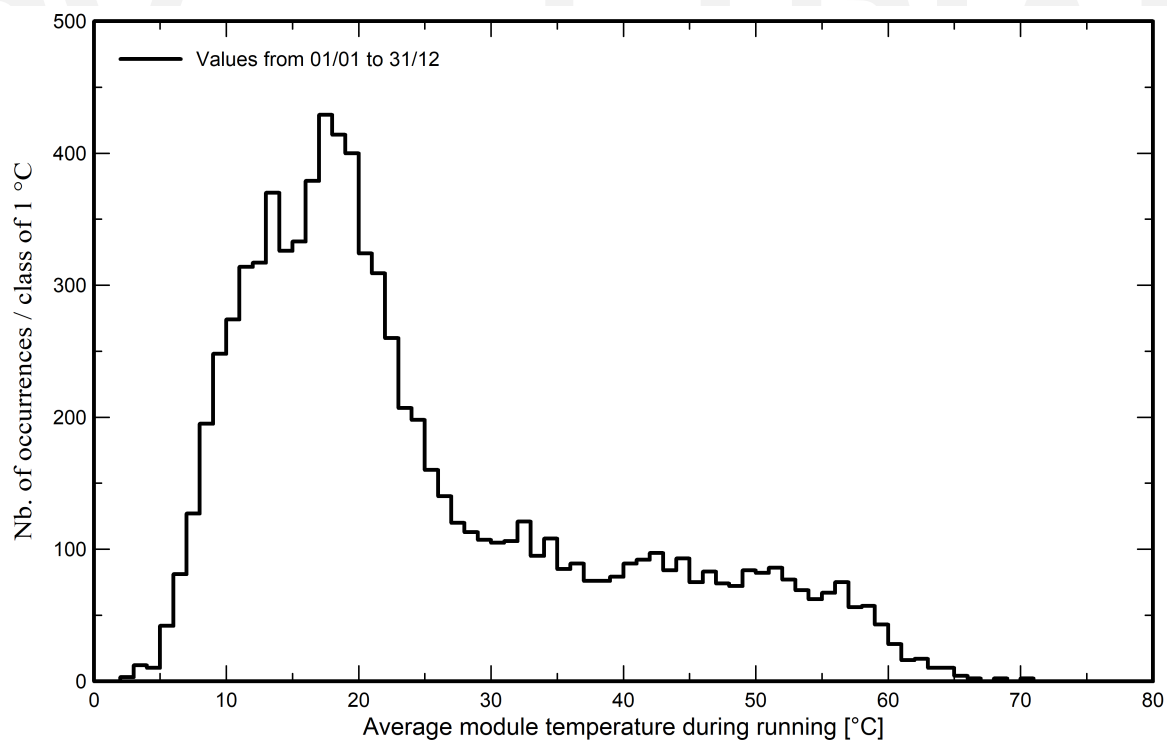
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Predef. graphs

Array Voltage Distribution



Array Temperature Distribution during running



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Cost of the system**Installation costs**

Item	Quantity units	Cost EUR	Total EUR
PV modules			
Mono 440 Wp Twin 144 half-cells	936	123.20	115 315.20
Supports for modules	936	17.60	16 473.60
Inverters			
350 kWac inverter, 12 MPPT	1	22 809.60	24 634.37
Other components			
Accessories, fasteners	1	8 236.80	8 236.80
Wiring	1	12 355.20	12 355.20
Combiner box	1	4 118.40	4 118.40
Monitoring system, display screen	1	4 118.40	4 118.40
Surge arrester	1	4 118.40	4 118.40
Studies and analysis			
Engineering	1	8 236.80	8 236.80
Permits and other adm. Fees	1	4 118.40	4 118.40
Installation			
Global installation cost per module	936	13.20	12 355.20
Global installation cost per inverter	1	3 801.60	4 105.73
Transport	1	4 118.40	4 118.40
Settings	1	4 118.40	4 118.40
Grid connection	1	4 118.40	4 118.40
Insurance			
Transport insurance	1	4 118.40	4 118.40
Liability insurance	1	2 059.20	2 059.20
Land costs			
Land preparation	1	4 118.40	4 118.40
		Total	240 837.70
		Depreciable asset	164 659.97

Operating costs

Item	Total EUR/year
Maintenance	
Cleaning	4 118.40
Total (OPEX)	4 118.40
Including inflation (2.00%)	5 003.31

System summary

Total installation cost	240 837.70 EUR
Operating costs (incl. inflation 2.00%/year)	5 003.31 EUR/year
Produced Energy	574 MWh/year
Cost of produced energy (LCOE)	0.0482 EUR/kWh

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Financial analysis**Simulation period**

Project lifetime 20 years Start year 2027

Income variation over time

Inflation 2.00 %/year
Module Degradation 0.50 %/year
Discount rate 6.50 %/year

Income dependent expenses

Income tax rate 0.00 %/year
Other income tax 0.00 %/year
Dividends 0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (EUR)	Depreciable (EUR)
PV modules				
Mono 440 Wp Twin 144 half-cells	Straight-line	20	0.00	115 315.20
Supports for modules	Straight-line	20	0.00	16 473.60
Inverters				
350 kWac inverter, 12 MPPT	Straight-line	20	0.00	24 634.37
Accessories, fasteners	Straight-line	20	0.00	8 236.80
		Total	0.00	164 659.97

Financing

Own funds 240 926.40 EUR

Electricity sale

Feed-in tariff 0.11000 EUR/kWh
Duration of tariff warranty 20 years
Annual connection tax 0.00 EUR/year
Annual tariff variation +3.0 %/year
Feed-in tariff decrease after warranty 0.00 %

Return on investment

Payback period 4.7 years
Net present value (NPV) 549 560.48 EUR
Internal rate of return (IRR) 26.66 %
Return on investment (ROI) 228.2 %



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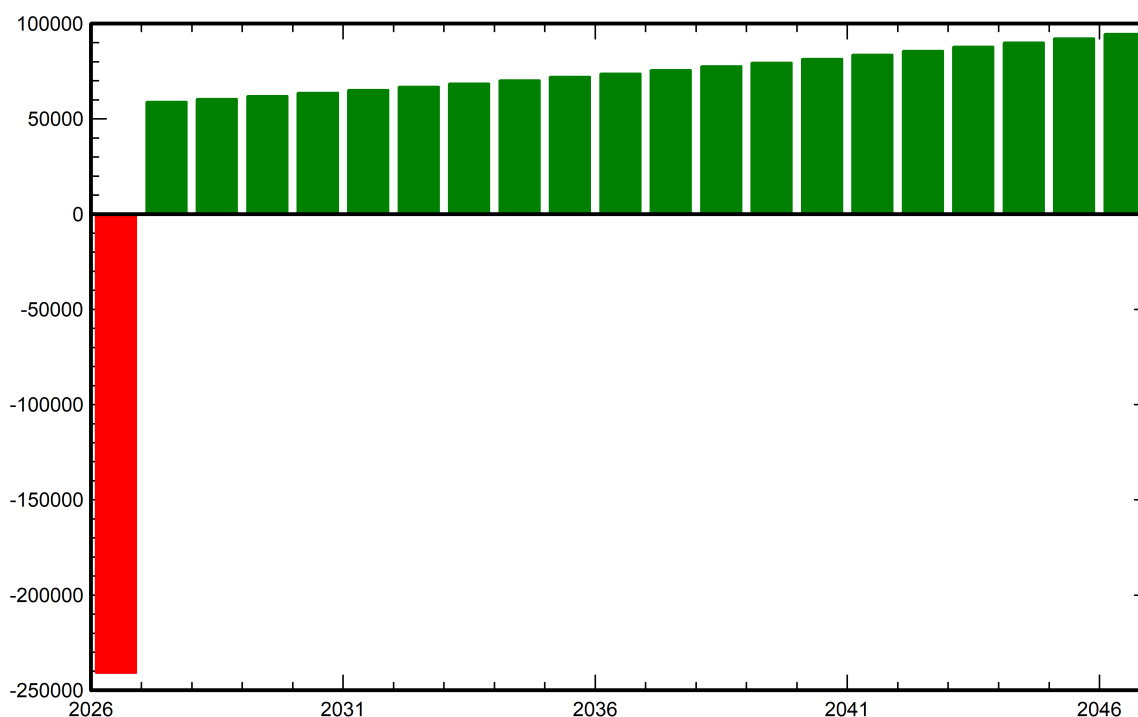
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Financial analysis

Detailed economic results (EUR)

Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Cumul. profit	% amorti.
0	0	240 926	0	0	0	0	0	-240 926	0.0%
1	63 139	0	4 118	8 233	50 788	0	59 021	-185 508	23.0%
2	64 708	0	4 201	8 233	52 274	0	60 507	-132 161	45.2%
3	66 316	0	4 285	8 233	53 798	0	62 031	-80 809	66.5%
4	67 964	0	4 370	8 233	55 360	0	63 593	-31 376	87.0%
5	69 653	0	4 458	8 233	56 962	0	65 195	16 208	106.8%
6	71 384	0	4 547	8 233	58 604	0	66 837	62 014	125.8%
7	73 158	0	4 638	8 233	60 287	0	68 520	106 107	144.1%
8	74 976	0	4 731	8 233	62 012	0	70 245	148 551	161.7%
9	76 839	0	4 825	8 233	63 780	0	72 013	189 408	178.7%
10	78 748	0	4 922	8 233	65 593	0	73 826	228 737	195.0%
11	80 705	0	5 020	8 233	67 452	0	75 685	266 595	210.7%
12	82 711	0	5 121	8 233	69 357	0	77 590	303 038	225.9%
13	84 766	0	5 223	8 233	71 310	0	79 543	338 118	240.4%
14	86 872	0	5 328	8 233	73 312	0	81 545	371 885	254.5%
15	89 031	0	5 434	8 233	75 364	0	83 597	404 390	267.9%
16	91 244	0	5 543	8 233	77 468	0	85 701	435 679	280.9%
17	93 511	0	5 654	8 233	79 624	0	87 857	465 798	293.4%
18	95 835	0	5 767	8 233	81 835	0	90 068	494 790	305.5%
19	98 216	0	5 882	8 233	84 101	0	92 334	522 697	317.1%
20	100 657	0	6 000	8 233	86 424	0	94 657	549 560	328.2%
Total	1 610 431	240 926	100 066	164 660	1 345 705	0	1 510 365	549 560	328.2%

Yearly net profit (EUR)

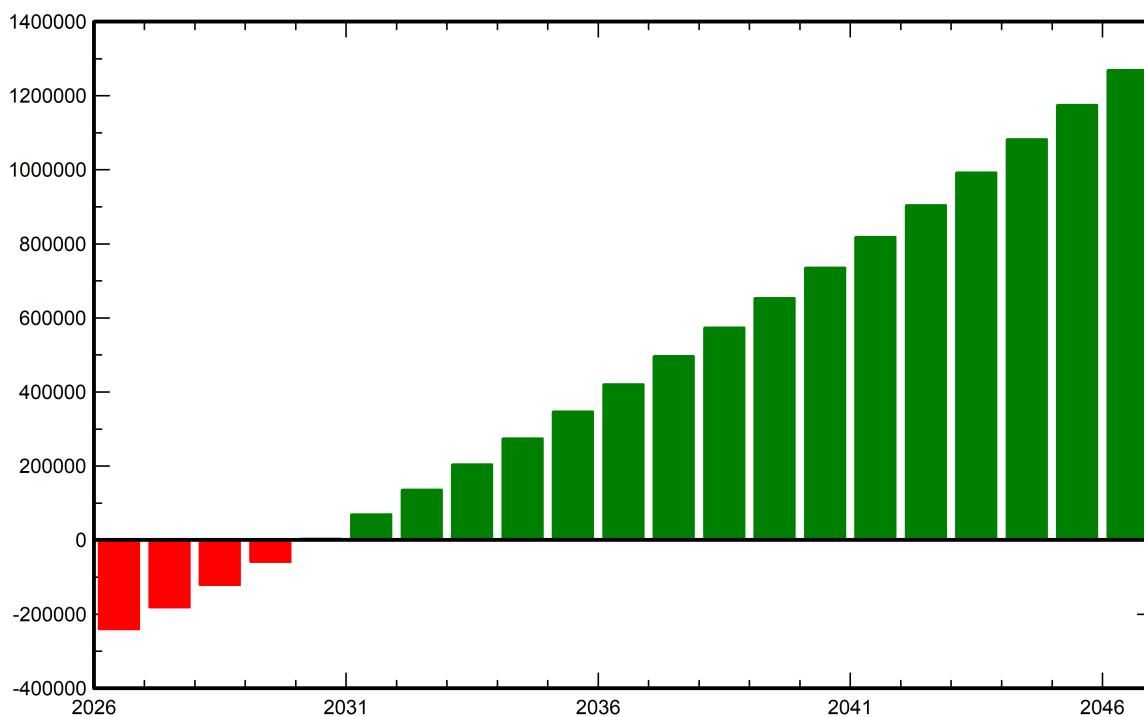




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Financial analysis
Cumulative cashflow (EUR)





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CO₂ Emission Balance

Total: 7224.5 tCO₂

Generated emissions

Total: 747.44 tCO₂

Source: Detailed calculation from table below

Replaced Emissions

Total: 8357.2 tCO₂

System production: 573.98 MWh/yr

Grid Lifecycle Emissions: 728 gCO₂/kWh

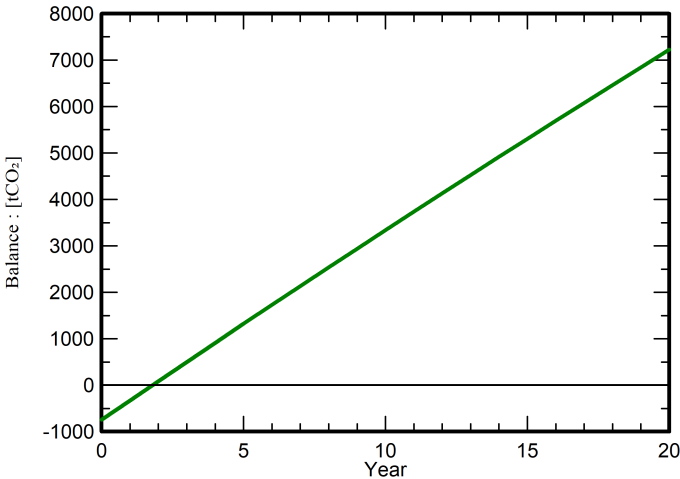
Source: IEA List

Country: Morocco

Lifetime: 20 years

Annual degradation: 0.5 %

Saved CO₂ Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[kgCO ₂]
Modules	1713 kgCO ₂ /kWp	412 kWp	705367
Supports	4.40 kgCO ₂ /kg	9360 kg	41205
Inverters	436 kgCO ₂ /units	2.00 units	872